Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [18CS3SPO3]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2021

DIGITAL ELECTRONICS

Time: 3 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. Perform following number conversion **3+3**

- **i.** Convert $(42AD)_{16}$ into a decimal number
- **ii.** Convert $(1010.011)_2$ into a decimal number

b. **i.** Convert $(367)_{10}$ into its Excess-3 code. **4+4**

ii. Add the BCD number 10010010 and 01011000.

2. a. Obtain the canonical sum of product form of the following function. **04**

F (A, B, C, D) = AB + ACD

b. Plot the logical expression F (A, B, C, D) = ABCD + A$\overline{BCD}$ + A$\overline{B}$C + AB on a four-variable Karnaugh map. Obtain the simplified expression **10**

3. a. Design a Full-adder circuit **07**

b. Implement the function F (A,B,C) = Σ (1,3,5,6) using decoder **07**

OR

4. a. Design a BCD-to-excess-3 Code Converter **09**

b. Implement the following function using a multiplexer. **05**

F(A, B, C) = (1, 3, 5, 6)

5. a. Differentiate combinational and sequential circuits **04**

b. Explain the working principle of SR FLIP-FLOP with truth table, and write its characteristics, excitation table **10**

OR

6. a. What is the race-around condition of a J-K flip-flop? How can it be avoided? **4**

b. Illustrate Conversion of an S-R Flip-flop to a D Flip-flop **10**

7. a. What are the four basic types of shift registers? Draw a block diagram for each of them. **06**

	b.	Design MOD-6 Asynchronous counter	**08**

OR

8.	**a.**	The content of a 4-bit register is initially 1011. The register is shifted 7 times to the right with the serial input being 1010110. What is the content of the register after each shift?	**05**
	b.	**i.** Differentiate synchronous and asynchronous counters	**3+6**
		ii. Design 2 bit synchronous counter	